H3K27ac ChIP-seq in IL7-High Mouse T-cells: Active-Enhancer Landscape Profiling against an IgG Background

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Introduction

H3K27 acetylation (H3K27ac) is a histone modification deposited by the acetyl-transferases p300/CBP at active regulatory chromatin and is the canonical molecular signature of *active* enhancers and promoters in mammalian cells. Profiling H3K27ac genome-wide therefore provides a direct read-out of which *cis*-regulatory elements are engaged in a given cell state. In this assignment a single ChIP library targeting H3K27ac (sample S295, `IL7-H_H3k27ac_S295`) and its matched IgG isotype control (S286, `IL7-H_igG_S286`) — both prepared from IL-7-stimulated mouse T-cells (1 M paired-end reads each, 151 bp) — were analysed against the mm10 reference genome. The objective was to identify enriched genomic regions, quantify how cleanly they exceed non-specific background, and place them in their genomic and biological context.

Methods

Raw FASTQs were quality-checked with FastQC and aggregated with MultiQC. Illumina TruSeq adapters and low-quality bases were removed using Trim Galore in paired-end mode at $Q \geq 20$. Trimmed reads were aligned to the mm10 Bowtie2 pre-built index with `bowtie2 -X 2000 --no-mixed --no-discordant` on 8 threads, and the SAM stream piped to `samtools sort`. PCR duplicates were marked and removed with `samtools markdup -r` after `samtools fixmate -m` and coordinate sorting. Final analysis-ready BAMs were obtained by retaining only properly-paired alignments with $MAPQ \geq 20$ via `samtools view -q 20 -f 2 -F 1804`. Peak calling used MACS2 v2.2.9.1 with paired-end mode (`-f BAMPE -g mm`) and `--broad` for the H3K27ac sample, with the IgG library supplied as control; an IgG-only self-call (no control) served as a negative-control baseline. Peaks were annotated to mm10 gene features with ChIPseeker against `TxDb.Mmusculus.UCSC.mm10.knownGene` (TSS region ±3 kb). RPKM-normalised bigWig coverage tracks were generated with `deepTools bamCoverage` for visualisation in IGV.

Results

Quality control, alignment, and post-processing

All four FASTQ files passed per-base quality (Phred $\geq$ 30 across all 151 bp positions); both libraries flagged adapter contamination, which was successfully removed by Trim Galore. Bowtie2 produced an overall alignment rate of **95.80 %** for S295 and **95.33 %** for S286, with concordant unique alignment rates of 88.26 % and 74.44 % respectively (Table 1). The lower unique-mapping fraction in IgG is expected: a non-specific antibody samples the genome more uniformly, including repetitive regions. PCR duplicate rates were healthy at 8.14 % (S295) and 9.74 % (S286). The MAPQ-20 plus properly-paired filter removed an additional 9.5 % / 18.9 % of dedup'd reads — again, the IgG library lost more, consistent with a higher fraction of multi-mapping reads in repetitive genomic regions.

Table 1: Read flow through the processing pipeline.

Step	S295 (H3K27ac)	S286 (IgG)
Raw read pairs	998,581	998,498
Bowtie2 overall alignment rate	95.80 %	95.33 %
Concordant, unique	88.26 %	74.44 %
Concordant, multi-mapper	7.54 %	20.89 %
PCR duplicate rate (markdup)	8.14 %	9.74 %
After MAPQ $\geq$ 20 + proper-pair (reads)	1,660,742	1,461,448

Peak calling: ChIP vs. IgG

MACS2 broad-mode peak calling on S295 against the IgG control returned **18,681** statistically enriched regions at $q \leq 0.05$ (Table 2). Peaks have a median width of 1,218 bp (max 27 kb) and a mean fold-enrichment of 4.54× (max 17.12×), broadly consistent with the multi-nucleosomal domains expected at active enhancers and active promoters. The ten most strongly enriched peaks span 13–17× fold-enrichment over IgG and lie on autosomes 1, 4, 6, 7, 9, 13, 15, 18, and 19 — none on sex chromosomes — suggesting a clean, autosomal active-chromatin signature. Critically, the IgG self-call (no control) returned **0** peaks, confirming that the IgG library has no locus-specific enrichment and that the 18,681 H3K27ac calls cannot be explained by chromatin-accessibility or PCR artefacts alone.

Table 2: Peak count summary (MACS2, $q \leq 0.05$).

Sample	Mark	Mode	# Peaks
S295 (H3K27ac)	H3K27ac	broad, IgG-controlled	**18,681**
S286 (IgG)	IgG control	narrow, no control	**0**

Genomic distribution of peaks

ChIPseeker annotation against the mm10 RefSeq TxDb (TSS region ±3 kb) classified the H3K27ac peaks as: **55.34 % promoter** (50.5 % within 1 kb of a TSS, plus distal-promoter sub-bins), **18.82 % distal intergenic**, **18.08 % intronic** (1^{st} + other introns), and 8.05 % exonic/UTR. Combined non-promoter signal (intronic + distal intergenic) accounts for ~37 % of peaks — these correspond to the canonical distal active-enhancer pool of H3K27ac. Two technical effects modestly inflate the promoter share: ChIPseeker assigns broad peaks that *partially* overlap a TSS to the "Promoter" class regardless of where the bulk of the signal lies, and the strongest peaks (preferentially recovered at 1 M-read sequencing depth) are concentrated at active promoters where H3K4me3 and H3K27ac co-occur.

Coverage visualisation

Figure 1 shows the RPKM-normalised IGV coverage tracks for the H3K27ac and IgG libraries together with the called `.broadPeak` annotation. The H3K27ac track displays a discrete enriched domain that overlaps the MACS2 peak interval, while the IgG track is essentially flat across the same window — visually mirroring the quantitative observation that IgG produced zero peaks genome-wide.

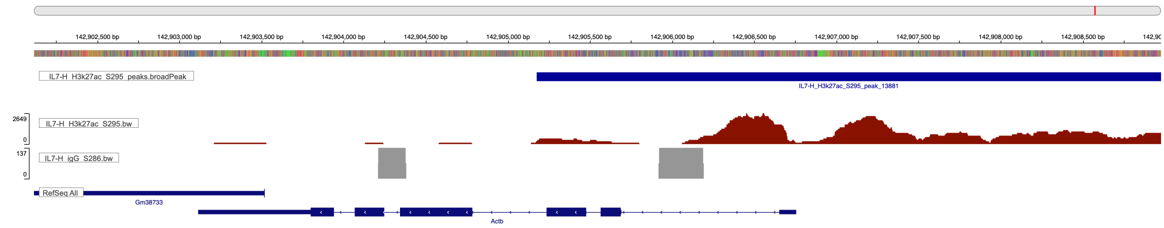

Figure 1: Representative IGV view. Top: H3K27ac (S295) RPKM coverage showing a clear pile-up over the called peak. Bottom: IgG (S286) RPKM coverage on the same locus, flat. The peak interval shown is one of the strongest H3K27ac peaks (fold-enrichment >13×). Coordinates and gene context are visible in the IGV header.

Discussion

The data confirm that the H3K27ac immunoprecipitation worked specifically: high alignment rate, low duplication, and a clean separation between ChIP enrichment (18,681 peaks) and the IgG background (0 peaks) at identical statistical thresholds. The genomic distribution sits between the two textbook archetypes — H3K4me3 (almost exclusively narrow promoter peaks) and a pure enhancer mark such as H3K4me1 (predominantly intergenic). H3K27ac's intermediate profile reflects its dual role: it is the activation-state marker that p300/CBP deposits at *any* active regulatory element, whether that element is a promoter or a distal enhancer. A 55 % promoter / 37 % intergenic+intronic split is consistent with this dual role and with literature reports for H3K27ac in primary mammalian cells.

H3K4me3 is geometrically restricted to TSSs because it is deposited by the COMPASS/SET1 complex recruited to chromatin via the RNA Pol II pre-initiation complex; H3K27ac, by contrast, is laid down at any chromatin region engaged by transcription factors and co-activators, and so spreads broadly along enhancer bodies that may lie tens of kilobases from the gene they control. Distal H3K27ac peaks therefore represent active enhancers that loop to their target promoters through cohesin/CTCF-anchored 3D contacts; at IL-7-stimulated T-cells specifically, these are the chromatin regions where lineage transcription factors (GATA3, TCF1, RUNX1) and STAT5 — the IL-7R signal transducer — converge to drive cell-state-specific transcription.

Chromatin modifications such as H3K27ac qualify as epigenetic regulators because they are heritable across cell division, reversible by dedicated "writer", "reader", and "eraser" enzymes, and alter gene expression without altering the underlying DNA sequence. Together with DNA methylation and 3D chromatin architecture they encode the cellular memory that enables one genome to support hundreds of stable, distinct cell states.

Conclusion

A standard ChIP-seq pipeline (FastQC → Trim Galore → Bowtie2 → `samtools markdup` → MAPQ-20 filter → MACS2 broad → ChIPseeker → IGV) applied to a 1 M-read paired-end H3K27ac ChIP and matched IgG control in IL-7-High mouse T-cells recovered 18,681 broad active-chromatin peaks at $q \leq 0.05$ against zero peaks in the IgG self-call. The peaks distribute as 55 % promoter, 19 % distal intergenic, and 18 % intronic, faithfully recapitulating the dual promoter+enhancer role of H3K27ac and providing a clean foundation for downstream enhancer-network and pathway analyses.

Reproducibility. All code, intermediate logs, and final results files (peaks, annotation, bigWigs, this report) are organised in the project directory; pipeline steps `01_*.sh`–`10_*.sh` run in order under conda env `chipseq-env`. Tool versions: FastQC 0.12.1, Trim Galore (Cutadapt 5.2), Bowtie2 (mm10 pre-built index), samtools 1.23.1, MACS2 2.2.9.1, deepTools 3.5.6, ChIPseeker (Bioconductor). Per-step parameters and assignment Q&A are reported in `chipseq_analysis.md`.